DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Design and Implementation of a 1-bit Full Adder using CMOS Logic Styles

Digital mini project: Static CMOS, PTL, Transmission Gate and GDI

MINI PROJECT 1 | DIGITAL DESIGN

Prepared for: VLSI Design Project Submission

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Academic status: ECE, 3rd semester

Academic design proposal. Numerical simulation results are not claimed until the designs are implemented and tested in a simulator.

Abstract

This project proposes transistor-level design and comparison of a 1-bit full adder using static CMOS, pass-transistor logic (PTL), transmission-gate logic and gate-diffusion-input (GDI). The same Boolean function and testbench will compare transistor count, propagation delay, average power and power-delay product (PDP). This report defines the architecture and verification plan; measured values must be added after simulation.

1. Problem statement

A full adder is a basic arithmetic building block used in adders, counters, address-generation units and arithmetic logic units. Different transistor-level logic styles trade off voltage swing, transistor count, speed, power and design complexity. A fair comparison requires identical input patterns, load assumptions, supply voltage and transistor models.

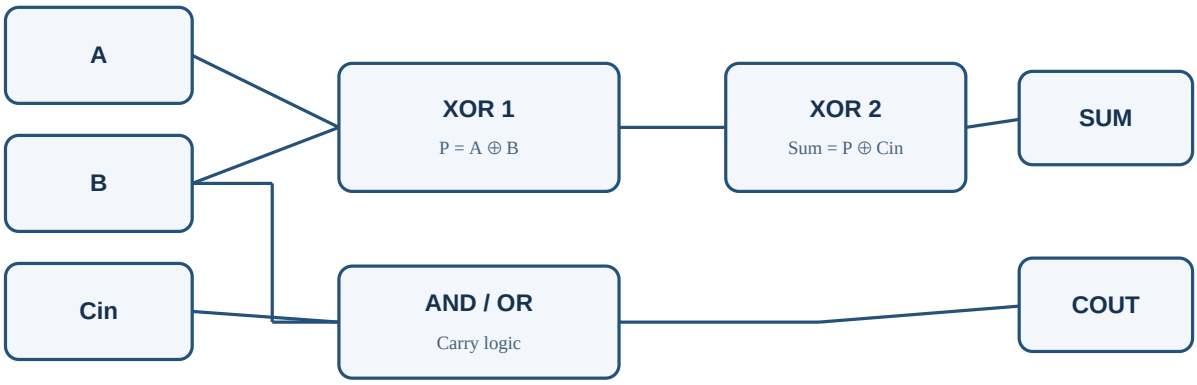
2. Objectives

• Derive and verify the full-adder Sum and Carry functions.
• Draw transistor-level implementations using four logic styles.
• Verify all eight input combinations in simulation.
• Compare transistor count, propagation delay, average power and PDP under common conditions.
• Explain design trade-offs without assuming a winner before measurement.

3. Theory and logic equations

Inputs are A, B and Cin. Outputs are Sum and Cout. Let P = A XOR B. Then Sum = P XOR Cin and Cout = (A AND B) OR (P AND Cin). These equations provide a common functional reference for all implementations.

Conceptual 1-bit full-adder signal flow



4. Logic styles

4.1 Static CMOS

Uses complementary pull-up and pull-down networks. It generally provides full-swing outputs and good noise margins, but may require more transistors than compact pass-based designs.

4.2 Pass-transistor logic (PTL)

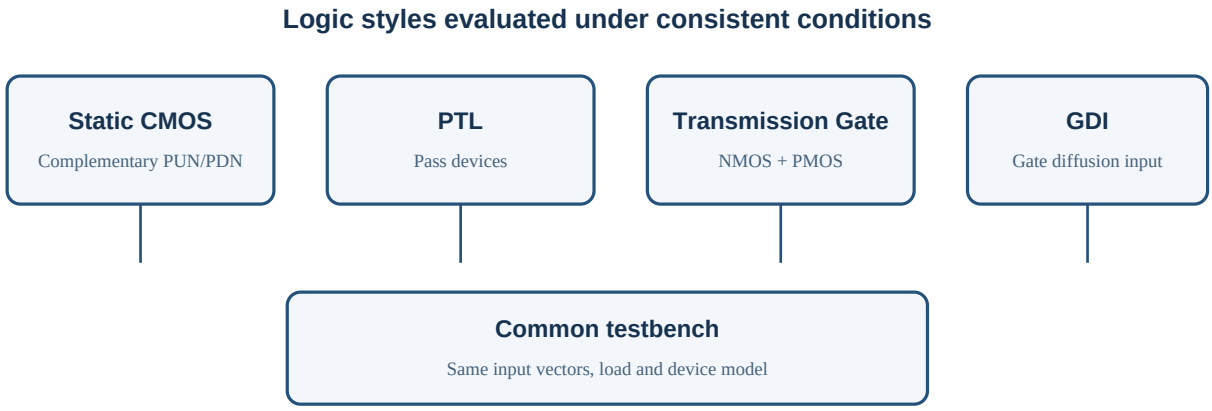
Uses transistors as controlled switches to pass logic levels. It can reduce device count, but threshold-voltage loss and output restoration may affect cascaded stages.

4.3 Transmission-gate logic

Uses parallel NMOS and PMOS devices driven by complementary controls. A transmission gate passes both logic levels more robustly than a single pass device, at the cost of complementary controls and additional devices.

4.4 Gate-diffusion-input (GDI)

GDI uses a cell structure with gate and diffusion inputs to implement compact Boolean functions. Practical implementation depends on the process, body connections and design-rule compatibility.



5. Truth table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

6. Proposed simulation methodology

1. Select a CMOS PDK or documented transistor model and record its version.
2. Implement four functionally equivalent circuits.
3. Apply all eight static input combinations and transient input sequences.
4. Use the same VDD, temperature, output load, input slew and measurement thresholds.
5. Measure worst-case propagation delay, average supply power and transistor count.

7. Performance metrics

Metric	Definition / method
Transistor count	Count all MOS devices; report auxiliary inverters separately.
Propagation delay	Measure input transition to 50% output transition; report worst relevant path.
Average power	Average VDD × supply current over a defined switching window.

Metric	Definition / method
PDP	Average power multiplied by propagation delay; state units and conditions.
Functionality	Compare simulated outputs against the eight-row truth table.

8. Results table to complete after simulation

Metric	Static CMOS	PTL	Transmission gate	GDI
Transistor count	TBD	TBD	TBD	TBD
Worst delay	TBD	TBD	TBD	TBD
Average power	TBD	TBD	TBD	TBD
PDP	TBD	TBD	TBD	TBD
All truth vectors pass?	TBD	TBD	TBD	TBD

9. Expected outcome and limitations

The expected outcome is a reproducible comparison showing how circuit style influences device count, delay and power. No style should be declared best until results are measured. The ranking may change with process model, supply voltage, sizing and load.

10. Conclusion

This mini project connects Boolean design with transistor-level VLSI implementation. Consistent test conditions and documented measurements can produce a useful comparison of full-adder architectures for low-power arithmetic circuits.

References

1. J. M. Rabaey et al., *Digital Integrated Circuits: A Design Perspective*.
2. N. H. E. Weste and D. Harris, *CMOS VLSI Design*.
3. Cite the selected PDK/model documentation and simulator manual in the final version.